

NAME :- M. SAHIL

Section :- "C"

Semester :- 6th

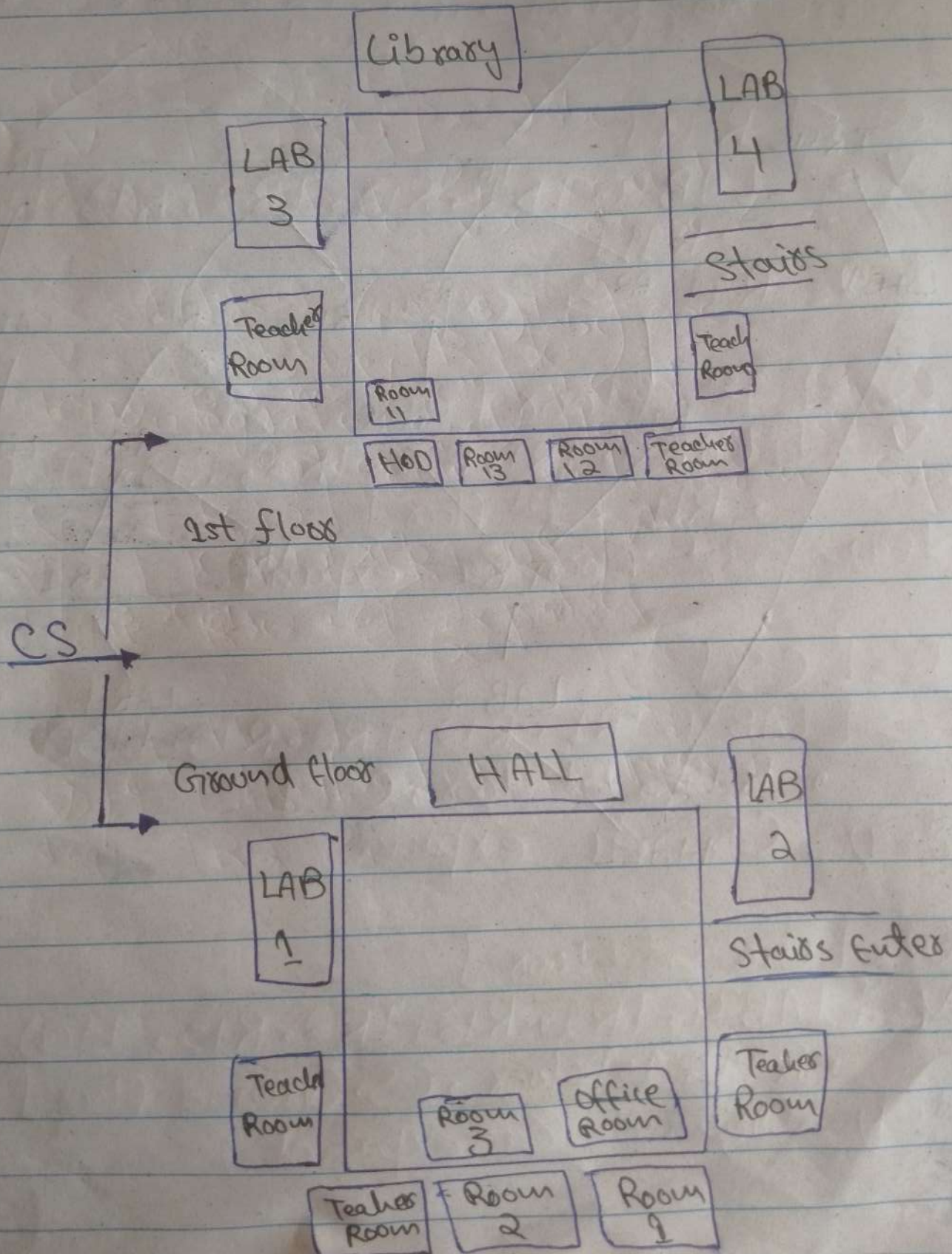
Roll no :- 23122137

Teacher :- Dr. B. M. M.

Task :- Assignment

Subject :- "AI"

Map of CS Department:-



Question NO 2:-

Transition table:-

CS $\rightarrow$ Ground floor, 1st floor

Ground floor $\rightarrow$ TR4, TR5

1st floor $\rightarrow$ HOD, TR3

TR4 $\rightarrow$ Lab 1, Hall

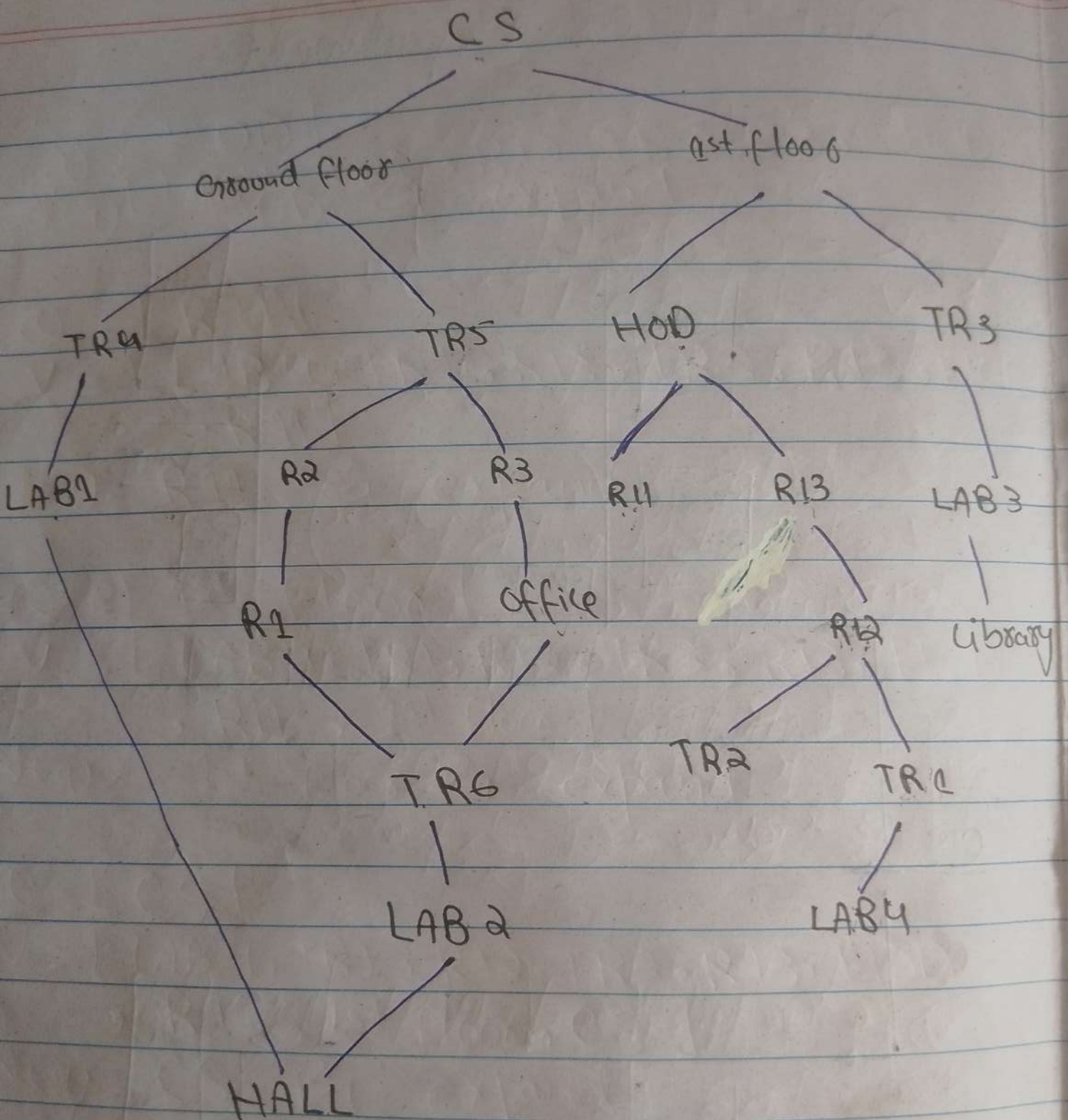
TR5 $\rightarrow$ R2, R1, R3, office, TR6

TR3 $\rightarrow$ Lab 3, Library

HOD $\rightarrow$ R11, R13, R12

R12 $\rightarrow$ TRA2, TR1, lab 4

TR6 $\rightarrow$ Lab 2



Question No 28-

Initial State = Lab 4

Goal State = Lab 2

State Space = CS Department

Action = Running

Transition table = Lab 4 to Lab 2

Path Cost = Cost 1

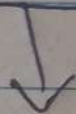
Question NO 4:-

Shortest Path :-

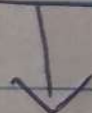
LAB 4



Stairs



Ground floor



LAB 2

LAB4 to LAB2

Initialization:-

$$Q = [LAB4]$$

$$V = [LAB4]$$

deq = LAB4 & expand

$$Q = [TR1]$$

$$V = [LAB4, TR1]$$

deq = TR1 & expand

$$Q = [R12]$$

$$V = [LAB4, TR1, R12]$$

deq = R12 & expand

$$Q = [TR2]$$

$$V = [LAB4, TR1, R12, TR2]$$

deq = TR2 & expand

$$Q = [R13]$$

$$V = [LAB4, TR1, R12, TR2, R13]$$

deq = R13 & expand

$$Q = [HOD]$$

$$V = [LAB4, TR1, R12, TR2, R13, HOD]$$

$$deq = HOD \text{ \& expand}$$

$$Q = [R11]$$

$$V = [LAB4, TR1, R12, TR2, R13, HOD, R11]$$

$$deq = R11 \text{ \& expand}$$

$$Q = [1^{st} \text{ Floor}]$$

$$V = [LAB4, TR1, R12, TR2, R13, HOD, R11, 1^{st} \text{ floor}]$$

$$deq = 1^{st} \text{ floor \& expand}$$

$$Q = [CS]$$

$$V = [LAB4, TR1, R12, TR2, R13, HOD, R11, 1^{st} \text{ floor}, CS]$$

$$deq = CS \text{ \& expand}$$

$$Q = [Ground \text{ Floor}]$$

$$V = [LAB4, TR1, R12, TR2, R13, HOD, R11, 1^{st} \text{ floor}, CS, \text{Ground Floor}]$$

$$deq = \text{Ground \& expand}$$

$$Q = [TR5]$$

$$V = [LAB4, TR1, R12, TR2, R13, HOD, R11, 1^{st} \text{ floor}, CS, \text{Ground floor}, TR5]$$

$$deg = TR5 \quad \&_1 \text{ expand}$$

$$Q = [R2]$$

$$V = [LAB4, TR1, R12, TR2, R13, HOD, R11, 1^{st} \text{ floor}, CS, \text{Ground floor}, TR5, R2]$$

$$deg = R2 \quad \&_1 \text{ expand}$$

$$Q = [R1]$$

$$V = [LAB4, TR1, R12, TR2, R13, HOD, R11, 1^{st} \text{ floor}, CS, \text{Ground floor}, TR5, R2, R1]$$

$$deg = R1 \quad \&_1 \text{ expand}$$

$$Q = [TR6]$$

$$V = [LAB4, TR1, R12, TR2, R13, HOD, R11, 1^{st} \text{ floor}, CS, \text{Ground floor}, TR5, R2, R1, TR6]$$

$$deg = TR6 \quad \&_1 \text{ expand}$$

Q = [LAB 2]

V = [LAB 4, TR 1, R 12, TR 2, R 13, HOO
R 11, 1st floor, CS, Ground floor
TR 5, R 2, R 1, TR 6, LAB 2]

deg = LAB 2 $\hookrightarrow$ expand - 100%